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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|
| 12 (a) (i) | ratio / dB = $10 \lg(P_1 / P_2)$<br>$24 = 10 \lg(P_1 / \{5.6 \times 10^{-19}\})$<br>$P_1 = 1.4 \times 10^{-16} \text{ W}$                                                                                                                                                                                                                                      | C1<br>C1<br>A1 | [3] |
| (ii)       | attenuation per unit length = $1 / L \times 10 \lg(P_1 / P_2)$<br>$1.9 = 1 / L \times 10 \lg(\{3.5 \times 10^{-3}\} / \{1.4 \times 10^{-16}\})$<br>$L = 1 \text{ km}$<br><i>or</i><br>attenuation = $10 \lg(\{3.5 \times 10^{-3}\} / \{5.6 \times 10^{-19}\})$<br>= 158 dB<br>attenuation along fibre = $(158 - 24)$<br>$L = (158 - 24) / 1.9 = 71 \text{ km}$ | C1<br>C1<br>A1 | [3] |
| (b)        | less attenuation (per unit length) / longer uninterrupted length of fibre                                                                                                                                                                                                                                                                                      | B1             | [1] |